

Molecular Subtypes and Diagnostic Biomarkers in Breast Oncology

A Review of Receptor Signaling Pathways and Genetic Risk Factors

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Abstract

Breast cancer is a heterogeneous group of malignant neoplasms defined by distinct molecular profiles, clinical behaviors, and therapeutic responses. This paper synthesizes current knowledge on Estrogen Receptor (ER), Progesterone Receptor (PR), Human Epidermal Growth Factor Receptor 2 (HER2), and germline BRCA gene mutations in diagnostic stratification.

1. Introduction and Tumor Heterogeneity

Carcinoma of the breast is among the most frequently diagnosed malignancies worldwide. Early anatomical classifications treated breast tumors as uniform pathological entities. However, gene expression profiling and immunohistochemistry have demonstrated substantial molecular heterogeneity, leading to distinct therapeutic targets and prognostic categories.

2. Molecular Subtypes and Receptor Signaling

Clinical oncology categorizes invasive breast carcinomas into four primary intrinsic subtypes based on hormone receptor and growth factor expression:

1. Luminal A (ER positive, PR positive, HER2 negative, low Ki-67 index): Characterized by favorable prognosis and high responsiveness to endocrine therapies such as tamoxifen and aromatase inhibitors. 2. Luminal B (ER positive, HER2 variable, high Ki-67 index): Exhibits higher proliferation rates and increased recurrence risk relative to Luminal A. 3. HER2-enriched (HER2 gene amplification/overexpression, ER negative, PR negative): Driven by ERBB2 gene amplification, targeted by monoclonal antibodies like trastuzumab. 4. Triple-Negative Breast Cancer (TNBC: ER negative, PR negative, HER2 negative): Lacks targeted hormonal or HER2 receptors, requiring cytotoxic chemotherapy regimens.

3. Genetic Risk Factors and DNA Repair Pathways

Germline mutations in BRCA1 and BRCA2 tumor suppressor genes account for a significant proportion of hereditary breast and ovarian cancer syndromes. BRCA1 and BRCA2 encode nuclear proteins essential for repairing double-strand DNA breaks via homologous recombination. Deficiencies in homologous recombination force cells to utilize error-prone pathways like non-homologous end joining, leading to genomic instability.

4. Advances in Non-Invasive Biomarker Detection

Mammographic screening remains foundational for early detection. However, novel diagnostic modalities such as liquid biopsies enable detection of circulating tumor DNA (ctDNA) and exosomes carrying tumor-specific mutations, offering opportunities for early intervention prior to clinical detection.

References

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